

 SECTION A: THEORY QUESTIONS (15 marks)

Q1. (a) Differentiate between Artificial Intelligence and Machine Learning with examples. (3 marks)

Answer: Artificial Intelligence (AI) is the broader concept of machines being able to carry out tasks in a way that we would consider “smart.”

Example: AI includes systems that can play chess, diagnose diseases, or provide customer support via chatbots.

Machine Learning (ML) is a subset of AI that focuses on algorithms and models that allow computers to learn from data and improve over time without being explicitly programmed.

Example: A spam filter that improves its detection as it processes more emails.

(b) Differentiate between Supervised and Unsupervised Learning. (3 marks)

Answer:

Feature	Supervised Learning	Unsupervised Learning
Data	Uses labeled data	Uses unlabeled data
Output	Predicts output for new data	Finds hidden patterns
Example	Regression, Classification	Clustering, Dimensionality Reduction
Use Case	Email spam detection	Market segmentation

Q2. What is the role of modeling in Machine Learning? Briefly explain the training process. (3 marks)

Answer:

Modeling is the process of creating a mathematical representation of a real-world process. In ML, the model learns from training data using an algorithm. During training:

- The model takes input data and maps it to output using a hypothesis.
- It compares predictions with actual output using a loss function.
- It updates its parameters (like weights) to reduce error, often using optimization methods like gradient descent.

Q3. Define the following evaluation metrics (1 mark each):

(i) MSE

(ii) RMSE

- (iii) R² Score
- (iv) Accuracy
- (v) F1 Score

Answer:

- (i) MSE (Mean Squared Error): Average of the squares of the errors between predicted and actual values.
- (ii) RMSE (Root Mean Squared Error): Square root of MSE; gives error in same unit as the target.
- (iii) R² Score (Coefficient of Determination): Proportion of the variance in the dependent variable that is predictable.
- (iv) Accuracy: Ratio of correctly predicted observations to the total observations.
- (v) F1 Score: Harmonic mean of precision and recall; useful when class distribution is imbalanced.

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SECTION B: APPLICATION/NUMERICAL (20 marks)

Q4. Given the following simple linear regression equation:

$$y = \beta_0 + \beta_1 x$$

Suppose we have points: (1, 2), (2, 3), (3, 4).

Find the best fitting line using the least squares method. (5 marks)

Answer:

Using formulas:

$$\beta_1 = \frac{\sum((x - \bar{x})(y - \bar{y}))}{\sum(x - \bar{x})^2}$$

$$\bar{x} = 2, \bar{y} = 3$$

$$\beta_1 = \frac{[(1-2)(2-3) + (2-2)(3-3) + (3-2)(4-3)]}{[(1-2)^2 + (2-2)^2 + (3-2)^2]} = \frac{(-1 \times -1 + 0 \times 0 + 1 \times 1)}{(1 + 0 + 1)} = \frac{1}{2} = 1$$

$$\beta_0 = \bar{y} - \beta_1 \bar{x} = 3 - 1 \times 2 = 1$$

$$\text{Equation: } y = 1 + 1x$$

Q5. Suppose we fit a regression model and obtained the following errors on 4 observations:

errors = [2, -1, -2, 1]

Compute the following: (5 marks)

- (a) MSE
- (b) RMSE
- (c) MAE

Answer:

$$(a) \text{ MSE} = \frac{(2^2 + (-1)^2 + (-2)^2 + 1^2)}{4} = \frac{(4+1+4+1)}{4} = \frac{10}{4} = 2.5$$

$$(b) \text{ RMSE} = \sqrt{2.5} \approx 1.58$$

$$(c) \text{ MAE} = \frac{(|2| + |-1| + |-2| + |1|)}{4} = \frac{(2 + 1 + 2 + 1)}{4} = \frac{6}{4} = 1.5$$

Q6. Explain the confusion matrix. Given the following matrix, compute accuracy, precision, recall, and F1 score. (6 marks)

Predicted:

	Positive	Negative
Actual +ve	40	10
Actual -ve	5	45

Answer: True Positives (TP) = 40

False Positives (FP) = 5

False Negatives (FN) = 10

True Negatives (TN) = 45

Accuracy = $(TP + TN) / \text{Total} = (40 + 45) / 100 = 85\%$

Precision = $TP / (TP + FP) = 40 / (40 + 5) = 40 / 45 \approx 0.89$

Recall = $TP / (TP + FN) = 40 / (40 + 10) = 40 / 50 = 0.8$

F1 Score = $2 \times (P \times R) / (P + R) = 2 \times (0.89 \times 0.8) / (0.89 + 0.8) \approx 0.842$

Q7. Briefly explain Gradient Descent and Loss Function. (4 marks)

Answer:

Gradient Descent is an optimization algorithm used to minimize a loss function by iteratively moving in the direction of the steepest descent. It updates the model's parameters (e.g., weights in linear regression) to reduce prediction error.

The Loss Function quantifies how well the model performs. For regression, common ones are MSE or MAE. The lower the loss, the better the model is performing.